



Modelling and simulating serpentine group behaviour in crowds using modified social force model

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ABSTRACT

Crowd dynamics has emerged as an important topic of investigation in recent years. Several studies have been conducted to better understand the underlying factors that can influence crowd dynamics and lead to risky situations. Particularly, field studies are essential for realistically capturing the dynamics. This paper attempts to uncover one intriguing behaviour observed in the world's largest mass religious gathering Kumbh Mela-2016, held in Ujjain, India. The authors define this behaviour, where groups attempt to overtake the others in the crowd, form flexible chains, and not allow non-group members to penetrate through them, as serpentine group behaviour. This behaviour has been modelled and simulated using a modified social force model which has been developed in this work. This can aid in the development of a realistic crowd simulation model that can handle such unique crowd behaviours and help manage large crowds and ensure safe movement.

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1. Introduction

In recent years, there has been a lot of research on pedestrian crowd dynamics [1–5]. Several experimental studies have been conducted to better understand the underlying factors that can influence crowd dynamics and potentially harm pedestrian safety [6–8]. However, some studies claim that experimental studies cannot recreate the original dynamics seen in the field, particularly pedestrians' emotions, motivation, and psychological factors [9,10]. This is even more difficult in the case of mass religious gatherings where people are highly motivated to reach a specific goal or destination within a specific time frame. To develop a crowd simulation model that can assist event planners and managers in such mass gatherings in better managing large crowds and ensuring safe movement, it is necessary to first understand the underlying factors that can pose serious threats to the safety and well-being of the participants. Furthermore, it is critical to study crowd dynamics in the field in order to capture them realistically.

Literature has long established the behaviour of pedestrians in large crowds [11–14]. While much research has been done on various crowd dynamics such as bottleneck [15,16], stop and go [17,18], self-organization [19,20], lane formation [21,22], synchronization of steps [23,24] and extreme dense crowd movements [3,25,26], a critical topic is social group behaviour [27]. A plethora of research has been conducted on studying the walking patterns of different social group sizes [28–30], the behaviour of social group members during evacuation [31,32], and the analysis of social group behaviour in different density conditions [33,34]. Furthermore, many have used force-based models [35,36], grid-based models [37,38], and heuristics-based models [39] to model social group behaviour. However, in the real-world scenario, other types

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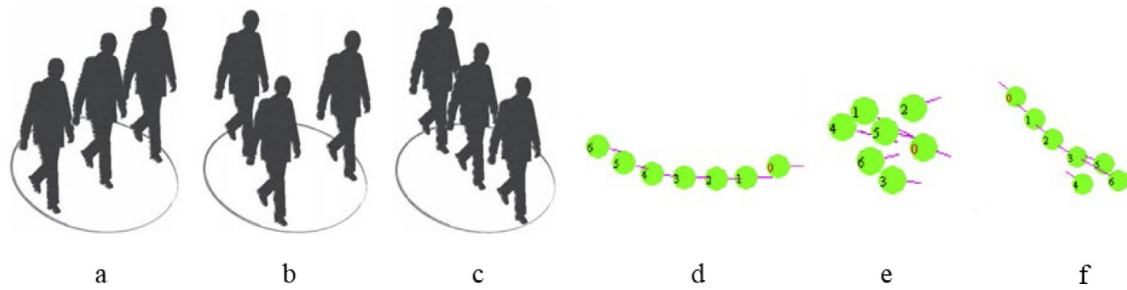


Fig. 1. Different walking patterns reported in literature (a) line abreast (b) V-like (c) riverlike (d) linear queue-like (e) leader-follower (f) mixed group structure.

of behaviours can be observed and must be addressed to improve the realism of crowd simulation models. The paper discusses one such behaviour discovered during the Kumbh Mela-2016 in Ujjain, India, which the authors define as the serpentine group behaviour.

What is a serpentine group?

Authors define serpentine group as “a group in which the members of the group (1) walk sequentially one after the other, holding hands or placing their hands on the shoulders of the previous member (basically in physical contact) forming a flexible chain (2) attempt to overtake other non-group members by sneaking through the crowd, and (3) do not allow non-members to penetrate through their group”. This behaviour ensures that the group members stay together and overtake others to reach the destination quickly. This behaviour in mass religious gatherings such as Kumbh Mela is induced by several factors such as the motivation to achieve the goal or reach the destination within a time frame, people viewing such events as a once-in-a-lifetime opportunity, a rare opportunity to be with their spiritual leaders, and the fear of losing the opportunity due to waiting and delay, which leads to exhibiting competitive behaviour [40].

Group shape and its walking patterns have been studied in the past to identify the group's underlying structure in crowd movement. The typical walking patterns of the groups studied so far are line abreast [41], V-like and riverlike patterns [6,29], queue-like pattern [31], linear, leader-follower and mixed group structure [42]. Fig. 1 shows the different walking patterns reported in the literature. Line abreast pattern is when group members walk alongside each other when there is sufficient free space, V-like is when the middle individual is slightly positioned behind the other group members in medium-density conditions, and riverlike is when the groups form a line in high-density condition for collision avoidance. Similarly, queue-like is when the group is self-organized for the evacuation process exhibiting cooperative behaviour. Although it seems to be similar to the context of serpentine groups, the queue-like pattern resulted in a positive effect on the crowd evacuation scenario studied by Xie et al. [31], while the serpentine chain is expected to have a negative effect on the overall crowd movement. A serpentine group's main motivation is to form continuous flexible chains in order to sneak through and overtake others in the crowd. Strong repulsive forces emerge when the distance between the serpentine group and the rest of the crowd decreases. As the number of serpentine groups grows, the crowd members become increasingly pressed against each other, with very little breathing space. This behaviour causes inadvertent body contact with other members of the crowd, resulting in a compressive force on them and pushing. As a result, the presence of a large serpentine chain increases the crowd risk [40]. Fig. 2 shows the observed serpentine group pattern in Kumbh Mela 2016.

This research aims to investigate the behaviour of such serpentine groups and their impact on crowd dynamics in general. The social force model was modified to account for the serpentine group's behaviour, and the parameters were calibrated using a field study conducted during the Kumbh Mela-2016 in Ujjain, India. The following sections discuss the empirical findings from the field study, the modified social force model, and the simulation results.

2. Empirical study

2.1. Data collection and extraction

The data for this study were gathered during the Kumbh Mela-2016, held in Ujjain, India. Kumbh Mela is one of the world's largest religious gatherings, with an estimated 75 million participants in the 2016 event. This event provided a unique opportunity to study the major critical dynamics of crowds, as described in [43]. Data correspond to events that took place on May 9, 2016, which was one of the event's most important days (known as Shahi Snan, the royal holy bath). The two main draws of Kumbh were (1) a dip in the river and (2) visits to temples. For this study, two stretches representing crowds near ghat section (meaning - a flight of steps leading down to a river) (G_1 “Ramghat”, G_2 “Choti Rapat”, where G represents ghat location) and two stretches representing crowds near temples (T_1 “Mahakal”, T_2 “Chardham”, where T represents temple location) were chosen. Fig. 3 depicts the study stretches with trap lengths and

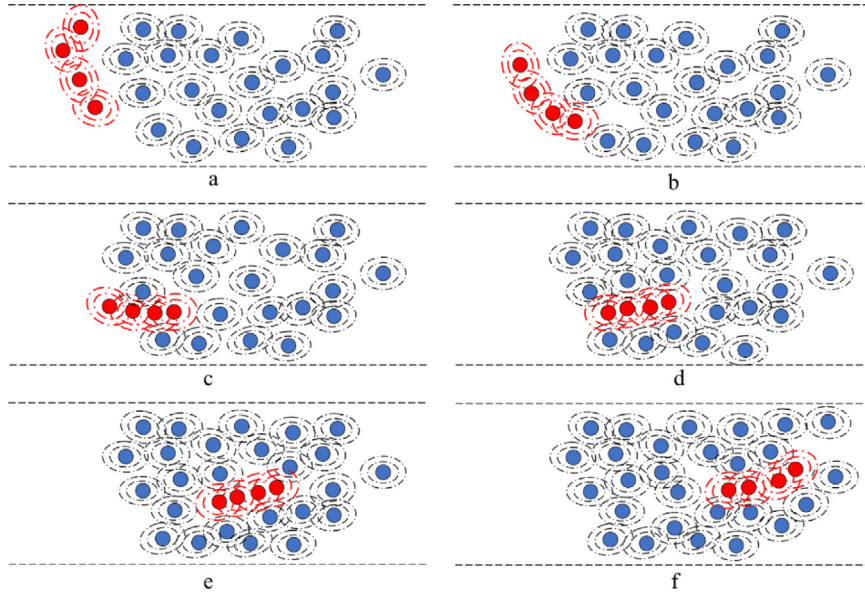


Fig. 2. Observed serpentine group pattern in Kumbh Mela 2016.

Table 1
Description of the study locations.

Location ID	Name	Description
G_1	Ramghat	Stretch leading to Ramghat where people go for a dip; unidirectional (at times people walk in both directions)
G_2	Choti Rapat	Stretch leading to Ramghat; unidirectional (at times people walk in both directions)
T_1	Mahakal	Stretch leading to Mahakal temple; unidirectional
T_2	Chardham	Stretch leading to Chardham temple; unidirectional

serpentine groups marked for all the locations. Wherever possible, data was collected using video cameras; otherwise, CCTV footage was gathered. The study locations are described in Table 1.

One-hour data was utilized for all the locations. Data extraction was done manually. Only groups that met the following criteria were chosen:

1. Walking sequentially one after the other with some form of physical contact (holding hands or shoulders) throughout the length of the study section considered.
2. Attempting to overtake other non-group members.
3. Refusing to allow others to penetrate through them.

Groups of sizes 2, 3, 4, and 5 were considered in this study, assuming that groups larger than 5 tend to split into subgroups [29]. Fig. 4 depicts the frequency of serpentine group sizes, location-wise. The number of group size 2 was more than the other group sizes, and it was observed that as the group size increases, the number of groups decreases. This could be because of the tendency of the groups to divide into subgroups. Average serpentine group speeds for all the locations were calculated using the following formula.

$$v_i = \frac{1}{n}(v_1 + \dots + v_n) \quad (1)$$

$$v_i = \frac{1}{n} \left(\frac{d}{(t_{1_{exit}} - t_{1_{entry}})} + \dots + \frac{d}{(t_{n_{exit}} - t_{n_{entry}})} \right) \quad (2)$$

where, v_i is the average speed of the group i of size n , v_1 and v_n is the speeds of the first and n th member of the group, d is the length of the trap marked, $t_{1_{entry}}$ and $t_{n_{entry}}$ is the time of the entry of the first and n th member of the group into the trap, $t_{1_{exit}}$ and $t_{n_{exit}}$ is the time of the exit of the first and n th member of the group from the trap. The serpentine speeds



Fig. 3. Snapshots of the study locations in Kumbh Mela 2016 with trap lengths (in orange lines) and serpentine groups (in white ellipses) marked.

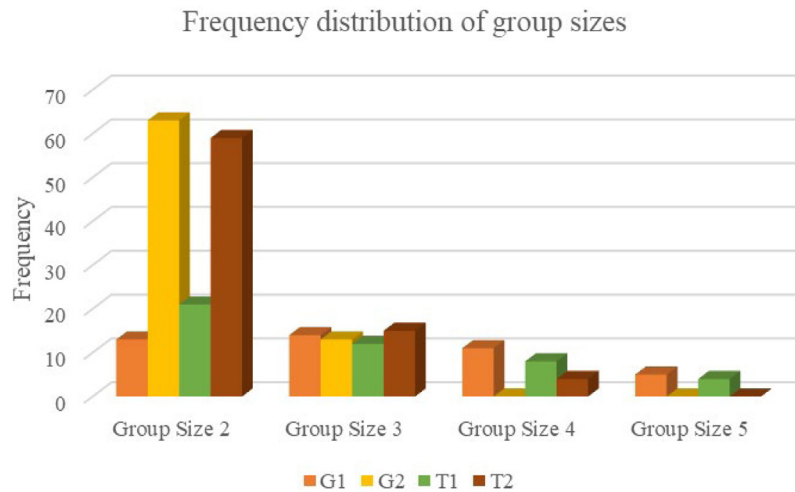


Fig. 4. Frequency distribution of group sizes observed in the study locations.

of all group sizes at T_1 were higher compared to other locations. The mean speeds in the ghat section were comparatively low. This could be due to the bidirectional movement seen in the ghat sections, even though it was supposed to be unidirectional. Table 2 summarizes the speeds (minimum, maximum, and average) and standard deviation. No noticeable trend between group size and average group speed as reported in the literature [44,45] was found in this case. Speeds up to 2.92 m/s were seen at some of the locations. The reason could be due to the induced motivation to move at their desired

Table 2

Minimum, maximum, and mean speeds of the group sizes observed in the study locations.

Group size	Population	Proportion (%)	Min speed (m/s)	Max speed (m/s)	Mean speed (m/s)	SD
G_1 (Total number of pedestrians = 943)						
2	13	2.8	0.64	1.66	1.07	0.34
3	14	4.5	0.42	1.36	0.96	0.27
4	11	4.7	0.54	1.11	0.81	0.18
5	5	2.7	0.57	1.39	1.03	0.34
G_2 (Total number of pedestrians = 3937)						
2	63	3.2	0.54	1.77	0.93	0.24
3	13	1.0	0.66	1.89	1.06	0.3
T_1 (Total number of pedestrians = 7418)						
2	21	0.6	0.68	2.28	1.66	0.38
3	12	0.5	0.89	2.92	1.53	0.53
4	8	0.4	0.91	2	1.38	0.38
5	4	0.3	0.86	2.31	1.34	0.65
T_2 (Total number of pedestrians = 631)						
2	59	18.7	0.34	1.12	0.65	0.16
3	15	7.1	0.39	0.94	0.68	0.16
4	4	2.5	0.47	1.03	0.63	0.26

speeds despite the high densities to reach their activity locations. There are also other possibilities, such as running or walking at higher speeds to reach out to their sub-group members who may be walking ahead of them, or time limits to perform certain activities. The authors believe that these are the likely causes of the observed high speeds.

The next section discusses the original social force model and the modified social force model for serpentine group behaviour developed by the authors.

3. Mathematical model

3.1. Helbing's social force model

The social force model developed by Helbing and Molnar [46] is widely used to model the behaviour of a pedestrian in a crowded environment. Pedestrian behaviour in a crowded environment is viewed as a combination of physical and psychological forces in this model, with the resultant force constituting the interaction of desired and interaction forces. Pedestrian behaviour is a complex phenomenon [47]. People in mass gatherings generally walk towards a particular goal or destination. A pedestrian i would prefer to walk with the desired speed v_i^0 in the desired direction $\vec{e}_i$ without any hindrance. This represents his motivation to reach the goal. However, in real grounds, pedestrian i 's interaction with fellow pedestrians, obstacles, and boundaries can affect his desired movement and his actual velocity $\vec{v}_i$ could be different from the desired velocity $\vec{v}_i^0$. Pedestrian i would still try to reach his desired velocity in the desired direction based on his personal driving force $\vec{f}_i^0$. This driving force which represents the motivation for i to act is given by,

$$\vec{f}_i^0 = m_i \frac{(v_i^0 \vec{e}_i) - \vec{v}_i}{\tau_i} \quad (3)$$

where, m_i is the mass of the pedestrian i and τ_i is the relaxation time.

As discussed earlier, in real grounds, pedestrian i 's motion can be disturbed due to another pedestrian, say j . Helbing and Molnar [46] talk about the private sphere that a pedestrian develops around himself, which, if intruded, would cause a repulsive effect. This interaction force between pedestrian i and pedestrian j is represented as,

$$\vec{f}_{ij}(\vec{r}_{ij}) = -\nabla_{\vec{r}_{ij}} V_{ij}[b(\vec{r}_{ij})] \quad (4)$$

where, $V_{ij}(b)$ is a monotonically decreasing repulsive potential, $\vec{r}_{ij}$ is the position of i with respect to j and b is the semi-minor axis of the equipotential lines that is assumed to be in the form of an ellipse (as shown in Fig. 5a).

The pedestrian i can also interact with walls and boundaries which can also cause a repulsive effect. This interaction force between pedestrian i and the boundary B is represented as,

$$\vec{f}_{iB}(\vec{r}_{iB}) = -\nabla_{\vec{r}_{iB}} U_{iB}[|\vec{r}_{iB}|] \quad (5)$$

where, $U_{iB}[|\vec{r}_{iB}|]$ is a monotonically decreasing repulsive potential, and $\vec{r}_{iB}$ is the position of i with respect to B .

The resultant force is represented as,

$$\vec{F}_i = m_i \frac{dv_i}{dt} = \vec{f}_i^0 + \sum_j \vec{f}_{ij}(\vec{r}_{ij}) + \sum_B \vec{f}_{iB}(\vec{r}_{iB}) \quad (6)$$

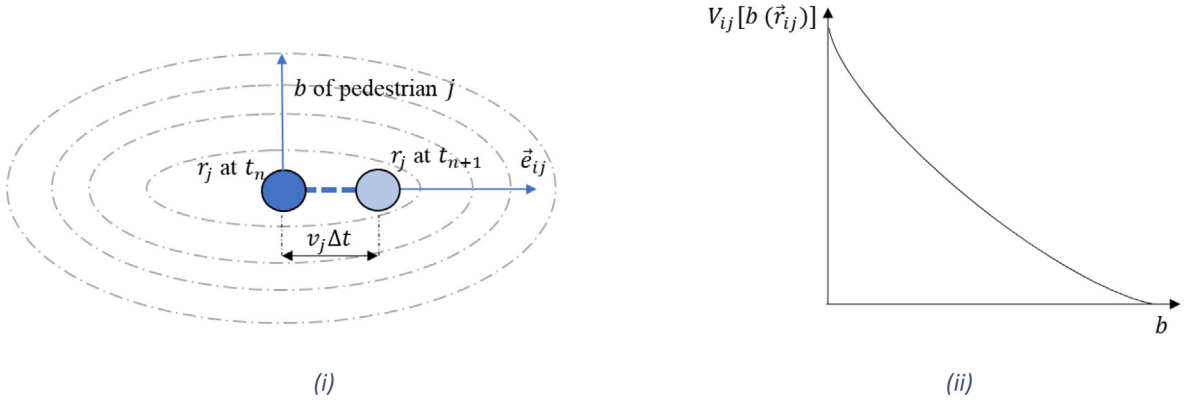


Fig. 5a. Equipotential lines in the form of an ellipse for a pedestrian j according to the Helbing's model (ii) Relationship between b and the repulsive potential.

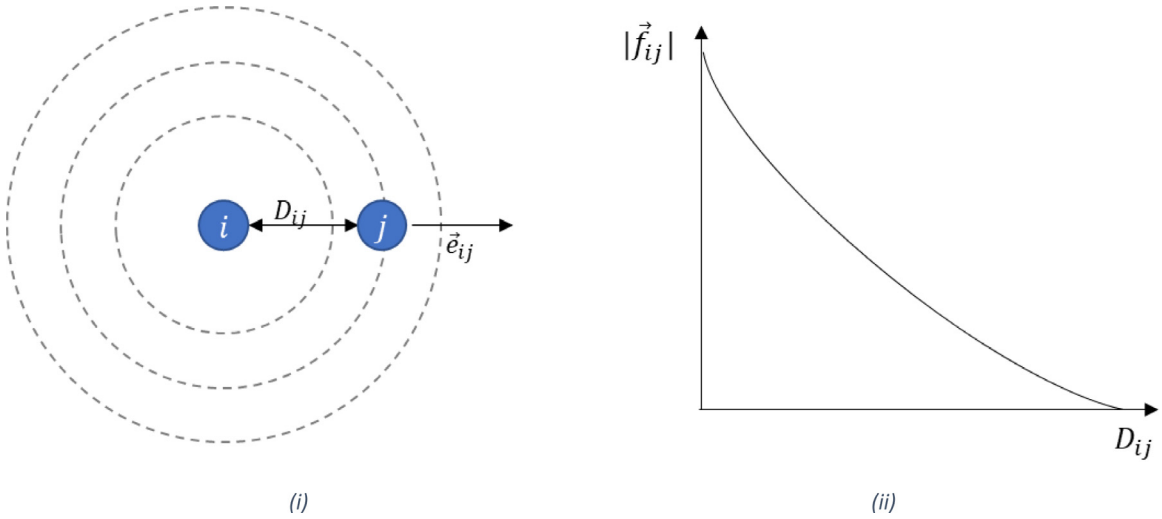


Fig. 5b. Equipotential lines in the form of a circle for a pedestrian j according to the Maury and Faure's model (ii) Relationship between b and the repulsive potential.

Pedestrian i can sometimes be attracted to other pedestrians (e.g. friends, street vendors) or boundaries (e.g. street art, window displays). Helbing and Molnar [46] have modelled this as a monotonically increasing attractive potential. However, this is omitted in this study for simplicity.

3.2. Overlapping constraint

Eqs. (7)–(10) are adopted from Maury and Faure [48]. The authors used another term in Eq. (4) to handle physical contact between individuals.

$$\vec{f}_{ij} = \left(-F \frac{-D_{ij}}{\delta} + \kappa (D_{ij}^-) \right) \vec{e}_{ij} \quad (7)$$

where, the coefficient $F = V_{ij}^0$ (homogeneous to a force) quantifies the social trend of i to keep apart from j . The coefficient δ (proxemic distance) is also involved in qualifying this trend to maintain a certain distance from neighbours. κ may be seen as a stiffness constant of individuals seen as deformable bodies. D_{ij} is the clear distance between the pedestrian i and j (as shown in Fig. 5b). $\vec{e}_{ij}$ is the direction in which the force is acting. D_{ij} and D_{ij}^- are represented as,

$$D_{ij} = \|\vec{x}_j - \vec{x}_i\| - r_i - r_j \quad (8)$$

$$D_{ij}^- = \begin{cases} D_{ij}, & \text{if } D_{ij} < 0 \\ 0, & \text{otherwise} \end{cases} \quad (9)$$

where, $\vec{x}_i$ and $\vec{x}_j$ are position vectors of pedestrian i and j .

The effects of obstacle/wall/boundary B are handled in similar way.

$$\vec{f}_{iB} = \left(-F_B \frac{-D_{iB}}{\delta} + \kappa (D_{iB}^-) \right) \vec{e}_{iB} \quad (10)$$

3.3. Modified social force model for serpentine group behaviour

The social force model developed by Helbing and Molnar [46] modelled pedestrian behaviour in panic and stressful situations without considering the psychological conditions. However, the social force model has served as a cornerstone for several researchers to build upon to improve the model, especially to account for group behaviour in a crowd. Xu and Duh [49] studied the bonding effects of pedestrians and their impact on crowd dynamics using the social force model by introducing bonding force and group attractiveness force. Šochman and Hogg [35] proposed a group inference task using the social force model and an agglomerative clustering algorithm for parameter estimation. Li et al. [50] proposed a grouping approach based on the social force model based on grid density and interpersonal relationships. Zhang et al. [51] proposed a two-layer social force model considering the inter-group attraction and leader–follower behaviour within groups during earthquake evacuation. Liu et al. [52] introduced a group force to quantify the relationship among the members in the group in a congestion situation and validated its effectiveness using a video data-driven model. Zhang et al. [53] used social comparison theory, a subjective self-evaluation method, in combination with crowd evacuation to evaluate the influence of group behaviour. Huang et al. [36] simulated social group behaviour in virtual geographic environments. Xie et al. [31] simulated group behaviour using a new group force Lennard-Jones potential which is used to describe the molecular interaction with another potential.

Many researchers have improved the social force model's performance in studying group behaviour. However, very few studies have included the phenomenon of groups overtaking in the social force model. Overtaking behaviour of groups can be observed in mass gatherings where people walk with a common goal in a high-density situation with desired velocities while remaining in close proximity with their group members. This behaviour can result in strong repulsive forces between pedestrians. Although there have been a few studies on overtaking behaviour in situations such as panic [54], congestion [55], and normal situations [56,57], no work has been done to study the effect of groups overtaking other pedestrians, particularly in large crowds, as per the authors' understanding. An interesting phenomenon observed in Kumbh Mela was a combination of groups attempting to overtake others, forming flexible chains and not allowing others to penetrate through them (serpentine group behaviour), which can cause severe disturbance to the equilibrium of the system. Therefore, the social force model was modified to account for this serpentine group behaviour.

When it comes to group behaviour, the social identity theory of leadership is widely applied to identify the social group leader. According to Hogg [58], the person who appears to have the most influence on the group members is generally chosen as the leader. In serpentine group case, the first person in the serpentine chain is considered as the leader due to the following reasons:

1. The first person is in the advantageous position of being the nearest to the destination.
2. It is assumed that the first person has the leadership skills to sneak through the available gaps in the crowd.

Consider a serpentine group i of n members walking towards their destination with i_1 being the leader and i_2 to i_n being the follower(s). For the leader, the goal is to reach the destination. For the follower(s), the ultimate goal is to reach the destination, but the immediate goal is to follow the previous member in the serpentine chain. According to Helbing and Molnar [46], the driving force to reach the desired destination for a pedestrian i is represented by Eq. (3).

In the case of a serpentine group, v_i^0 and $\vec{e}_i$ are,

1. For the leader i_1 , $v_{i_1}^0$ and $e_{i_1}^0$
2. For the follower(s), $v_{i_2}^0$ to $v_{i_n}^0$ and $e_{i_2}^0$ to $e_{i_n}^0$, respectively.

To maintain the serpentine group cohesion, a serpentine harmony coefficient α is introduced. α is assumed to be the same for all the followers for simplicity. This coefficient explains the degree to which a pedestrian would want to follow his previous member in the serpentine chain. A high value of α indicates more group cohesion. To reach the destination/goal, a pedestrian would try to take the shortest possible way. This way will usually have the shape of a polygon with edges $\vec{r}_i^1, \dots, \vec{r}_i^n := \vec{r}_i^0$ [46]. $\vec{r}_{goal}^k$ is the next edge of this polygon to reach goal and $\vec{r}_{i_1}^k$ is the next edge of this polygon to reach position of group leader, similarly, $\vec{r}_{i_{n-1}}^k$ is the next edge of this polygon to reach position of $(n-1)^{th}$ member. The desired direction $\vec{e}_i^0$ for the leader and the follower(s) is,

1. For the leader i_1 ,

$$e_{i_1}^0 = \frac{\vec{r}_{goal}^k - \vec{r}_{i_1}(t)}{|\vec{r}_{goal}^k - \vec{r}_{i_1}(t)|} \quad (11)$$

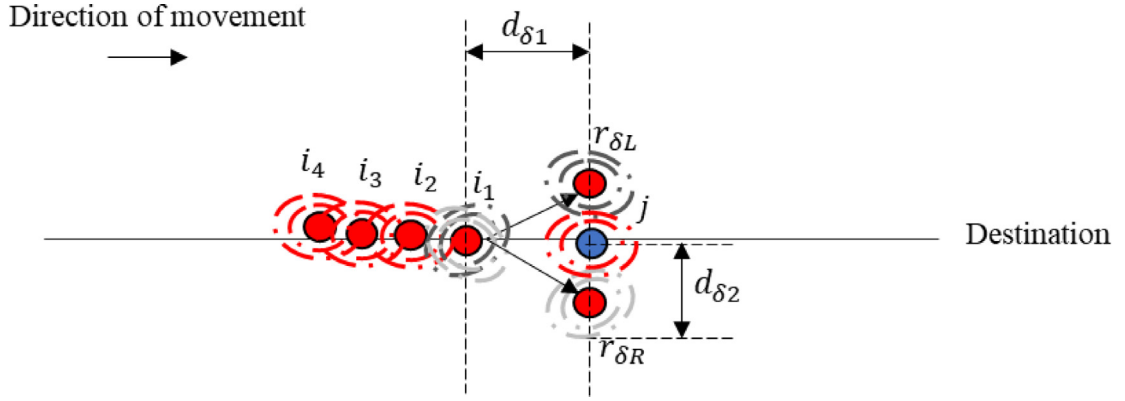


Fig. 6. Overtaking behaviour of a serpentine group (blue represents isolated pedestrians and red represents serpentine group).

2. For the followers,

$$\vec{e}_{(i_2)}^0 = \alpha \frac{\vec{r}_{i_1}^k - \vec{r}_{i_2}(t)}{|\vec{r}_{i_1}^k - \vec{r}_{i_2}(t)|} + (1 - \alpha) \frac{\vec{r}_{goal}^k - \vec{r}_{i_2}(t)}{|\vec{r}_{goal}^k - \vec{r}_{i_2}(t)|} \quad (12)$$

$$\vec{e}_{(i_n)}^0 = \alpha \frac{\vec{r}_{i_{n-1}}^k - \vec{r}_{i_n}(t)}{|\vec{r}_{i_{n-1}}^k - \vec{r}_{i_n}(t)|} + (1 - \alpha) \frac{\vec{r}_{goal}^k - \vec{r}_{i_n}(t)}{|\vec{r}_{goal}^k - \vec{r}_{i_n}(t)|} \quad (13)$$

Substituting Eqs. (11)–(13) in Eq. (3) and arriving at the serpentine group members' individual driving force, the average driving force of a member in a serpentine group towards the goal is,

$$\vec{F}_{average,i}^0 = \frac{1}{n} (\vec{f}_{i_1}^0 + \dots + \vec{f}_{i_n}^0) \quad (14)$$

Overtaking behaviour contributes more to the interaction forces than the desired forces. If a pedestrian j intrudes into the personal sphere of pedestrian i , then a strong repulsive force acts on pedestrian i . When a pedestrian tries to avoid another pedestrian, he can either overtake him on his left or right [59]. Adopting this, when a serpentine group tries to overtake pedestrian j sneaking through the small gap available near pedestrian j , the position that the leader i_1 first takes is given by,

1. If i_1 overtakes on j 's left,

$$\vec{r}_{\delta L} = \vec{r}_j + d_{\delta 1} \vec{e}_{i_1} + d_{\delta 2} \vec{e}_{i_1}^\perp \quad (15)$$

2. If i_1 overtakes on j 's right,

$$\vec{r}_{\delta R} = \vec{r}_j + d_{\delta 1} \vec{e}_{i_1} - d_{\delta 2} \vec{e}_{i_1}^\perp \quad (16)$$

where, $d_{\delta 1}$ and $d_{\delta 2}$ are the longitudinal and lateral distance between pedestrian j , $\vec{r}_{\delta L}$ and $\vec{r}_{\delta R}$ are the next desired position of i_1 on the left and right side of pedestrian j respectively, and $\vec{e}_{i_1}$ and $\vec{e}_{i_1}^\perp$ are the desired direction and unit normal to the desired direction. This is explained in Fig. 6.

The repulsive force exerted on pedestrian i_n of serpentine group by pedestrian j can be expressed by Eq. (7). The repulsive force from the boundary on i_n is given by Eq. (10). The follower(s) position and the repulsive force are going to be the same as that of the previous member in the serpentine chain.

Therefore, the average repulsive force on a member of serpentine group i of size n from pedestrian j is,

$$\vec{F}_{average,ij}^0 = \frac{1}{n} (\vec{f}_{i_1j}(\vec{r}_{i_1j}) + \dots + \vec{f}_{i_nj}(\vec{r}_{i_nj})) \quad (17)$$

Therefore, the average repulsive force on a member of serpentine group i of size n from the boundary/obstacle B is,

$$\vec{F}_{average,iB}^0 = \frac{1}{n} (\vec{f}_{i_1j}(\vec{r}_{i_1j}) + \dots + \vec{f}_{i_nj}(\vec{r}_{i_nj})) \quad (18)$$

The resultant force acting on the serpentine group i of n members is,

$$\vec{F}_{average,i} = m_i \frac{dv_i}{dt} = \vec{F}_{average,i}^0 + \sum_j \vec{F}_{average,ij}^0(\vec{r}_{ij}) + \sum_B \vec{F}_{average,iB}^0(\vec{r}_{iB}) \quad (19)$$

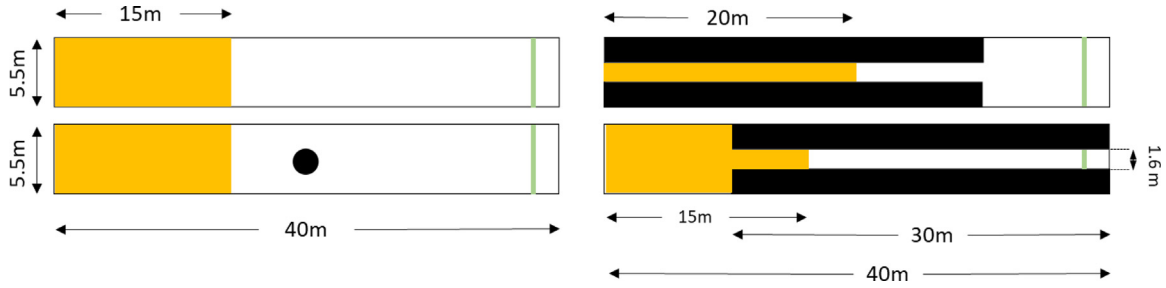


Fig. 7. Geometries considered for scenario analysis: Corridor (top-left), Corridor with an obstacle in the middle (bottom-left), Bottleneck (top-right) and Reverse bottleneck (bottom-right). The initialization region is marked in yellow, and the destination is marked in green.

The repulsive force between the members of the serpentine group is assumed to be nil as the group members choose to stay together in physical contact and not break the chain.

4. Simulation

4.1. Scenarios considered

The simulation of pedestrians was done separately, with the ghat and temple data captured/extracted as mentioned in the previous sections. Cromosim, a Crowd Motion Simulation tool developed by Sylvain Faure, is used for the simulation (<https://pypi.org/project/cromosim/>). The social force model implemented in the tool was modified to account for the serpentine group behaviour. A few scenarios with different geometries (Fig. 7) observed in Kumbh were simulated to study the effect of serpentine groups. The scenarios considered were:

1. Corridor (40 m length * 5.5 m width)
2. Corridor with an obstacle in the middle (40 m length * 5.5 m width, 1 m radius obstacle)
3. Bottleneck (40 m length, 5.5 m wide channel narrowing down to 1.6 m)
4. Reverse bottleneck (40 m length, 1.6 m narrow channel widening up to 5.5 m)

For each scenario, serpentine harmony coefficient, α , was set to (0, 0.2, 0.4, 0.6, 0.8, 1) to understand how the group cohesion influences the group speed and the time taken for them to reach the destination. In addition, the following compositions were tried to further understand their behaviour.

- GC1. Only isolated pedestrians (no serpentine group)
- GC2. Isolated pedestrians + serpentine groups of size 2 only.
- GC3. Isolated pedestrians + serpentine groups of size 3 only.
- GC4. Isolated pedestrians + serpentine groups of size 4 only.
- GC5. Isolated pedestrians + serpentine groups of size 5 only.
- GC6. Isolated pedestrians + Kumbh composition (G_1 and T_1 for ghat and temple, respectively)

All the compositions mentioned above were simulated with values taken from G_1 and T_1 . Therefore, total 288 simulations ($2 * 4 * 6 * 6$) were performed. These simulations were further repeated five times with different pedestrian location initialization (within the yellow region). Fig. 8 shows the simulation flowchart.

4.2. Results and discussion

About 200 pedestrians were generated for all the scenarios except the reverse bottleneck, where only 100 pedestrians were generated. The reason was that at high density and less width of the bottleneck channel, the pedestrians were found to move out of the domain. The initial speeds of the different group sizes for ghat and temple were obtained from G_1 and T_1 speeds summarized in Table 2. The initial speed of isolated pedestrians was assumed to be 0.9 m/s. V_{inj} and U_{inB} were assumed as $2.1 \text{ m}^2\text{s}^{-2}$ and $10 \text{ m}^2\text{s}^{-2}$ from Helbing and Molnar [46]. The assumptions from Maury and Faure [48] are listed below.

$$m_i = 80 \text{ kg } \forall i, \quad \tau_i = 0.5 \text{ s } \forall i, \quad F = 2000 \text{ N}, \quad d_{s1} = 0.5 \text{ m}, \\ \delta = 0.08 \text{ m}, \quad \kappa = 120000 \text{ kg s}^{-2}, \quad F_B = 2000 \text{ N}, \quad d_{s2} = 0.5 \text{ m},$$

Fig. 9 shows the screenshots of the simulation for different geometries for GC6 with α value 0.6.

Fig. 10 shows the comparison of the average speeds of all the pedestrians across the length of the section. For all scenarios, it was observed that the speeds of serpentine groups, in general, were higher than isolated pedestrians' speeds.

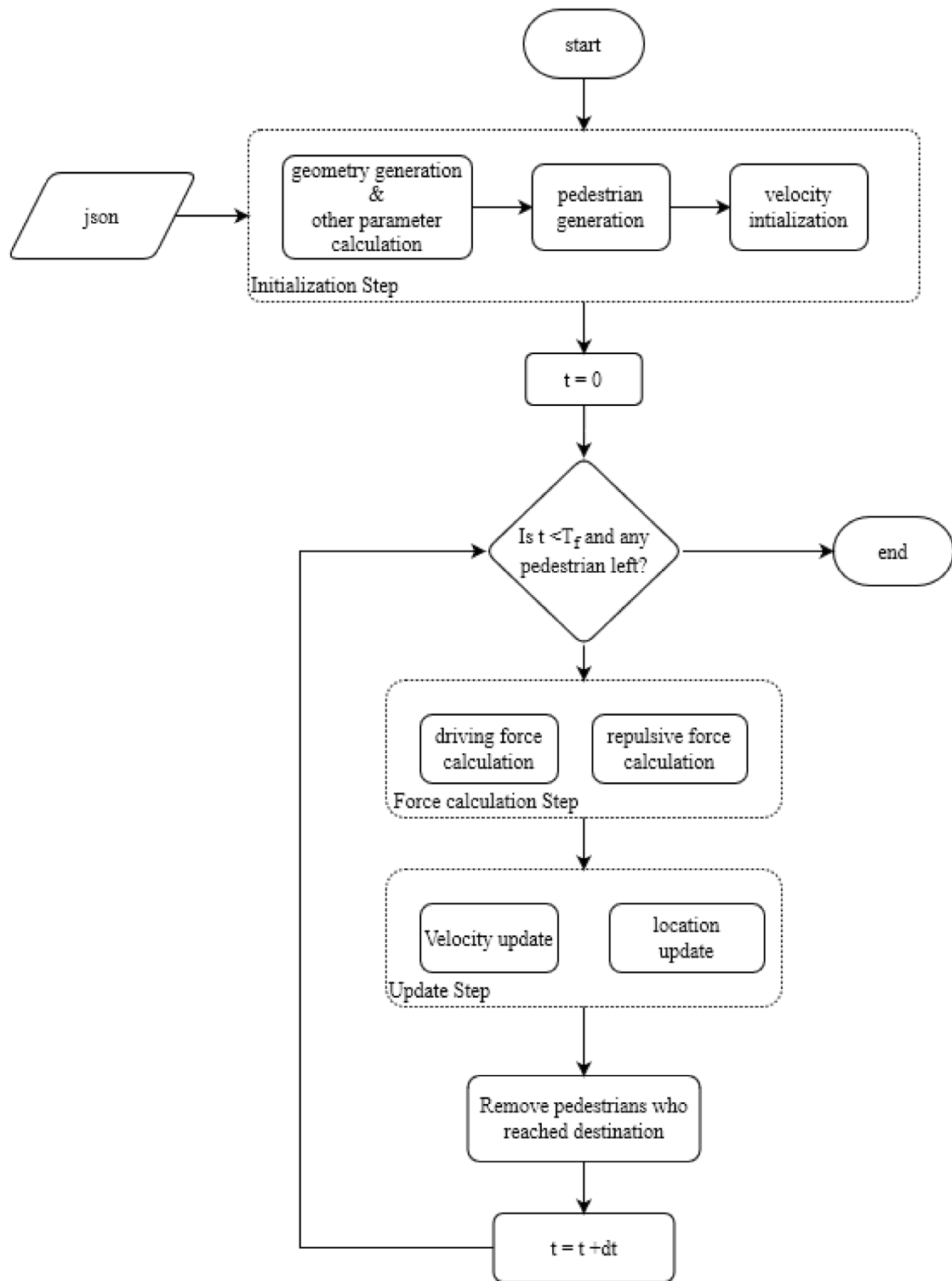


Fig. 8. Simulation flowchart.

While comparing the speeds of isolated pedestrians across the different group compositions, the average speeds in GC1 were less. It was also clear that α does not have any influence on the isolated pedestrians. For the serpentine groups, in most of the cases, average speeds were more at α equal to 0 and less at α equal to 1. In all the cases (except GC3), as α



Fig. 9. Screenshots of simulation for different geometries for GC6 with α value 0.6: Corridor (top-left), Corridor with an obstacle in the middle (bottom-left), Bottleneck (top-right) and Reverse bottleneck (bottom-right).

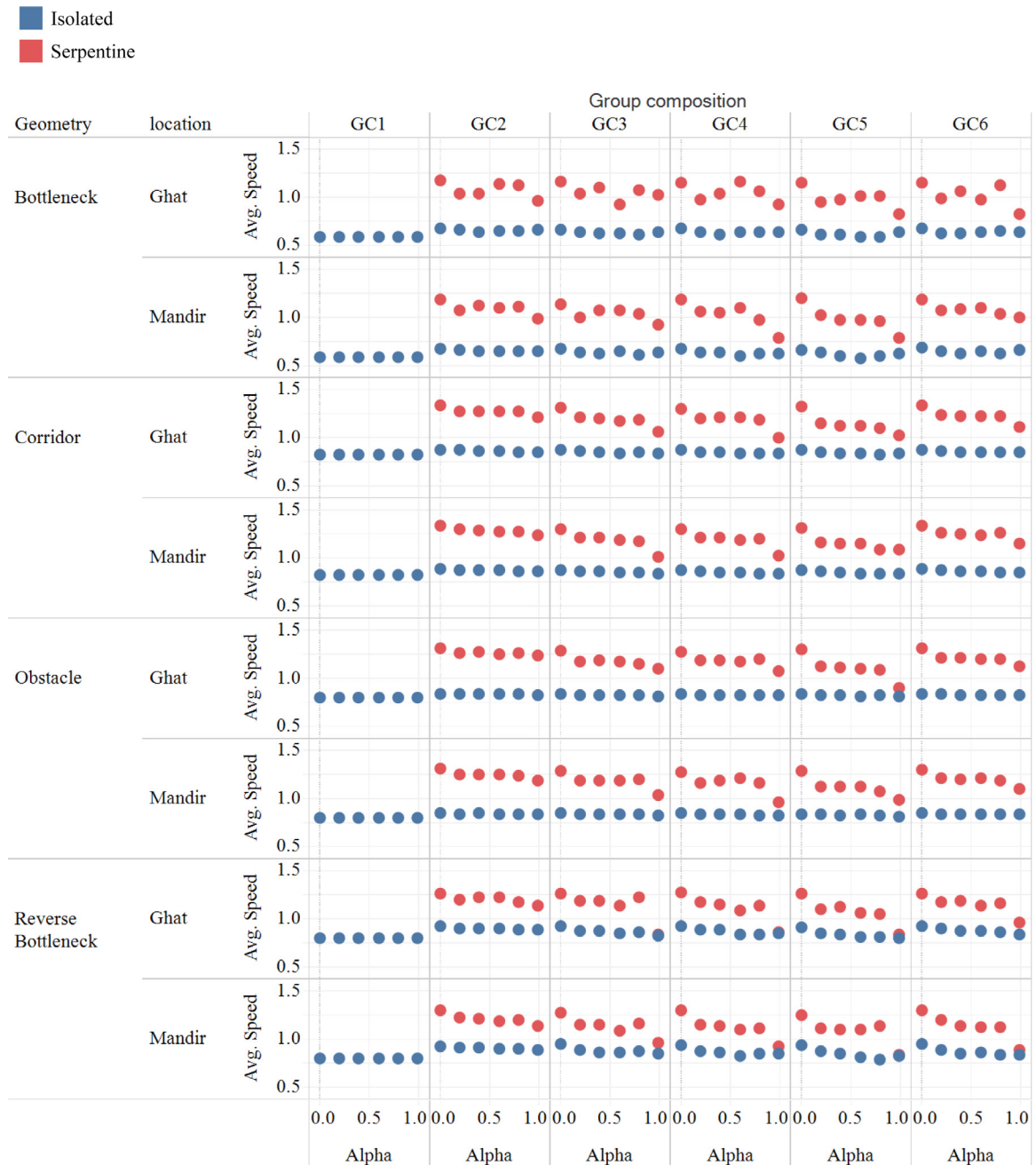


Fig. 10. Comparison of average speed across different scenarios.

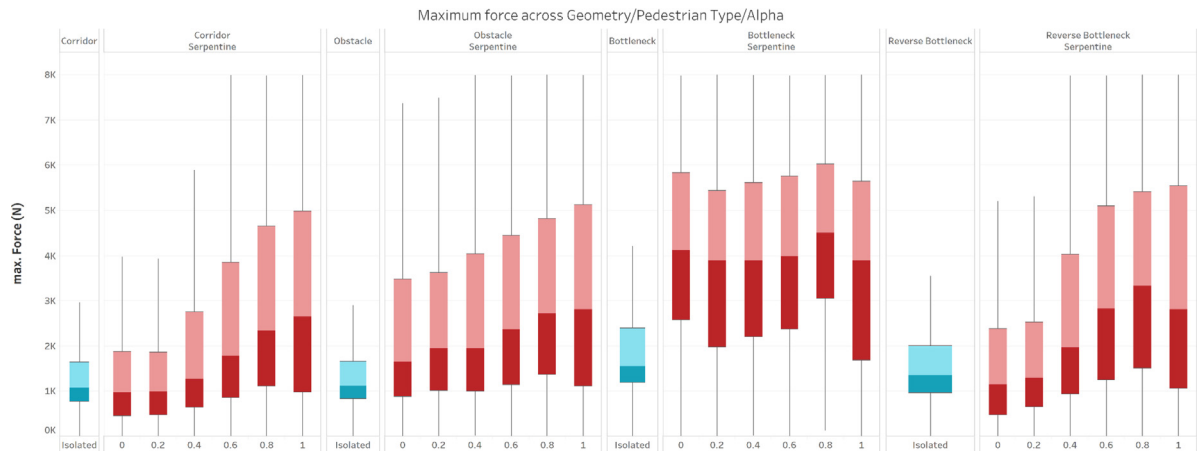


Fig. 11. Maximum force exerted on isolated pedestrians and pedestrians in serpentine groups for different α values and geometries, for GC6.

increased, average speeds decreased. This indicates that if group cohesiveness is more, the group members reach the goal with lower average speeds. The average speeds across the serpentine group compositions were almost the same, with the speeds being slightly high for GC6. While comparing across the different geometries, it was observed that the bottleneck had more variation to the α values. This indicated that the speeds of serpentine groups with high group cohesiveness were not affected much but posed a negative impact on the isolated pedestrians whose average speeds were comparatively less. From the average speed values, it was understood that the average time taken for isolated pedestrians to reach the destination across all the scenarios was more compared to serpentine groups that walked with much higher speeds and reached the destination quickly. The average time taken across the configurations with serpentine groups was almost the same but slightly less for GC6, complimenting their average speed values.

Figs. 11 and 12 show the maximum and average repulsive force acted on isolated pedestrians and serpentine groups for different geometries and α values for the GC6. For all geometries and α values, serpentine groups were found to exert greater forces than isolated pedestrians. However, the average force was higher for isolated pedestrians. This could be because isolated pedestrians spent more time in the region of interest because they walked at a slower pace than serpentine groups. In addition, they encountered numerous serpentine groups attempting to overtake them. As a result, while serpentine groups received the most force, the average force in the region studied was higher for isolated pedestrians, putting them at higher risk. Among the different geometries, the maximum force on serpentine groups exerted was high for the bottleneck section, and the average force on isolated pedestrians exerted was high for the same bottleneck section. There is evidence of physical forces beyond 4500 N acting on people in crowd crushes that led to bent of steel railings [60]. Literature has reported that the main cause of death in such crowd crushes is asphyxiation due to the physical forces, where people are squeezed and suffocated to death. Reports of people being lifted to more 10 ft above the ground due to such high physical forces were also observed [60]. In their simulation, Yu and Johansson [61] assumed the maximum repulsive force for dense areas as 160 N, which is very less than what was observed in GC6. In Kumbh Mela, where people are spiritually motivated, a small psychological trigger can result in such unanticipated events. With a simulation density of only 4 ped/m², the maximum repulsive force observed was nearly 6000 N which can potentially lead to bending of steel railings, as seen in previous incidents. Therefore, the authors strongly believe that the modified model developed explains both physical and psychological forces. Contrary to the physical forces, the psychological forces may not be equal for both isolated pedestrians and serpentine groups, which is reflected in the simulation also. The difference is that the serpentine groups are overtaking, and the isolated pedestrians are being overtaken. This strongly suggests that the average force exerted on isolated pedestrians is high, as they are subjected to more psychological forces because of the serpentine group behaviour than the serpentine groups themselves. Therefore, it is strongly believed that the psychological force acting on the spiritually motivated pedestrians, combined with physical forces due to serpentine group behaviour, is the primary cause of such high forces in the medium-density situation considered.

4.3. Comparison of speeds and forces across different scenarios

The 1-minute average speeds of isolated pedestrians of one of the sections in Kumbh Mela 2016, other than the sections considered in the paper, was extracted. This was again averaged over an hour. This average speed of 0.72 m/s was used as the initial isolated pedestrian speed instead of the assumed one. Table 3 summarizes the average speeds, average force, and maximum force for all the alpha values and geometries with the isolated pedestrian speed. The same pattern of serpentine groups having higher average speeds than the isolated pedestrians was observed. Also, the maximum force exerted on serpentine groups is higher than the isolated pedestrians while the average force exerted is lower than that of isolated pedestrians.

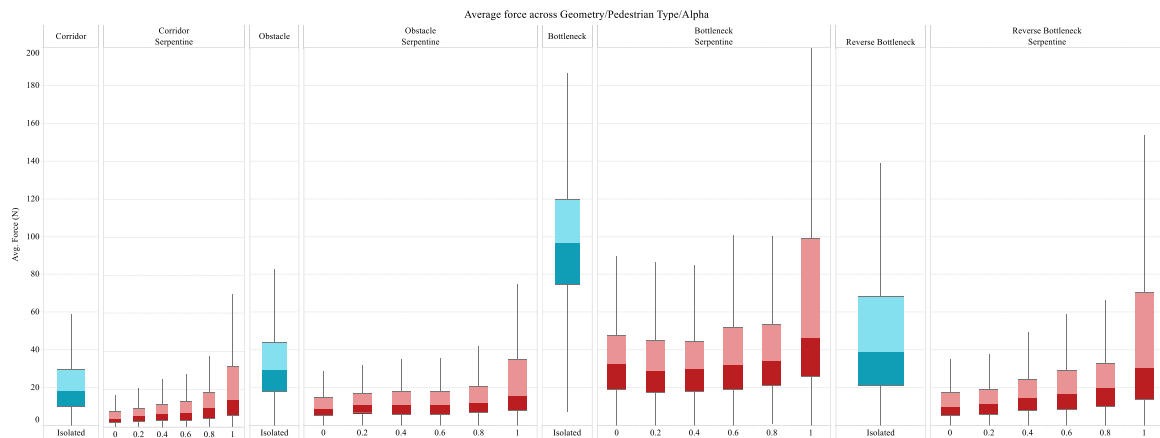


Fig. 12. Average force exerted on isolated pedestrians and serpentine groups for different α values and geometries, for GC6.

Table 3

The average speeds, average force and maximum force obtained from the isolated pedestrian speed.

Type	Isolated	Serpentine					
Alpha		0	0.2	0.4	0.6	0.8	1
Average speed (m/s)							
Bottleneck	0.62	1.12	0.97	1.04	1.06	1.04	1.03
Corridor	0.73	1.24	1.16	1.15	1.17	1.18	1.22
Corridor with obstacle	0.71	1.22	1.16	1.16	1.16	1.19	1.2
Reverse bottleneck	0.76	1.1	1.09	1.15	1.05	1.08	1.23
Average force (N)							
Bottleneck	92.8	38.41	31.22	40.82	37.99	38.7	42.54
Corridor	21.75	5.31	7.02	5.82	13.81	16.2	17.17
Corridor with obstacle	29.2	12.11	12.9	10.55	11.92	11.95	42.94
Reverse bottleneck	218.63	17.45	22.01	19.64	18.29	32.1	150.43
Maximum force (N)							
Bottleneck	2020.1	6426.0	5551.7	7694.4	6910.7	7502.3	7738.9
Corridor	1416.9	1577.8	2727.9	1573.9	5051.9	5692.4	3641.2
Corridor with obstacle	1364.3	2828.3	3399.9	3177.8	3647.1	4010.4	4920.0
Reverse bottleneck	6062.5	3514.1	3483.6	4723.6	6663.9	5663.4	8275.1

5. Summary and conclusion

In mass religious gatherings like Kumbh Mela, the motivation levels of people are high due to spirituality. They are highly motivated to reach the destination at the earliest, and this sense of urgency has led to exhibiting behaviours that can threaten the safety of crowds. Previous studies in Kumbh Mela have found that induced motivation is one of the driving factors that compel people to walk for longer distances [62,63]. This paper studied an interesting phenomenon that was witnessed in Kumbh Mela-2016 called serpentine group behaviour. The overtaking behaviour of serpentine groups, coupled with sneaking through the crowd forming flexible chains, exerted social force on the non-group members in the crowd. A modified social force model was developed to account for this serpentine group behaviour. Different geometries were simulated to understand the serpentine groups' behaviour in different situations. From the simulation results, it was observed that the average speeds of serpentine groups were more than isolated pedestrians as they try to overtake others and reach the destination before others. The induced motivation has acted as a catalyst on the people's minds to exhibit such behaviours. It can also be inferred from the average speeds that the time taken for them to reach would be much lesser than isolated pedestrians. The important findings of this study are:

1. The average speeds of serpentine groups are much higher than isolated pedestrians indicating that the serpentine groups reach the destination or their goal much earlier than the isolated pedestrians.
2. The average force exerted on isolated pedestrians is high, as they are subjected to more psychological forces because of the serpentine group behaviour than the serpentine groups themselves.

This serpentine movement restricts the differential movement of other pedestrians in the crowd creating more chances for a risky situation. The leader and follower(s) of a serpentine group ensure that they stay together (exerting attractive

force between the group members) as well as try to overtake others (exerting repulsive force on non-group members). When the density is less, attractive force dominated, which is not of much concern. However, as the density increases, repulsive force dominates, which led to such high repulsive forces. This can create inadvertent body contacts leading to pushing and other domino effects that have swayed many lives in several incidents, including 1971 Ibrox Park soccer stadium, 2010 Love Parade, and 2015 Hajj. This serpentine group behaviour exhibited in Kumbh Mela-2016 has exerted high physical and psychological forces on isolated pedestrians as observed, leading to potential crowd crushing and compressive asphyxia. However, the interaction between serpentine groups is not studied, which is a limitation.

This study tries to uncover the underlying factors that can pose serious threats to the safety and well-being of the participants but ignored in the literature. This can aid in the development of a realistic crowd simulation model that can handle such unusual crowd behaviours, allowing event planners and managers of such mass religious gatherings to better manage large crowds and ensure safe movement. The social force model is used in this study because of its simplicity and good fit in describing pedestrian behaviour. However, there are other pedestrian models that can explain pedestrian behaviour more realistically but not investigated in this study. This could be a future objective. As it is difficult to measure the psychological forces acting on an individual, the validation of forces was not done, which is a limitation. Further, the parameters assumed in the simulation can be altered to study how the model performs with different values. Furthermore, the geometries considered in the study have been observed in Kumbh Mela-2016. However, there can be other geometries and scenarios based on the region/city/event where serpentine group behaviour can be observed and have a more negative impact on the pedestrians, which can also be studied in the future.

CRedit authorship contribution statement

Gayathri Harihara Subramanian: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Nipun Choubey:** Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Ashish Verma:** Conceptualization, Data curation, Funding acquisition, Investigation, Project administration, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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